

Yigit Onat

yionat@gmail.com · [TODO: phone] · [TODO: city, country]
[TODO: SSRN author page URL — exists after P2] · [TODO: GitHub, if the code is to be shown]

Education

[TODO: years] MSc, [TODO: exact degree title] — [TODO: institution].

Thesis: [TODO: title].

Advisor: [TODO: name] — this person is referee slot 1].

[TODO: grade / distinction, if favourable; omit if not]

[TODO: years] BSc, [TODO: degree] — [TODO: institution].

Professional experience

[TODO: years] [TODO: role] — [TODO: organisation].

[TODO: 2–3 lines. Emphasise: systems where correctness mattered, independent technical judgement, scale. This is what referee slot 2 will be asked to confirm.]

[TODO: years] [TODO: role] — [TODO: organisation]. [TODO: 1–2 lines]

Independent research (2025–2026)

Built and operate an end-to-end quantitative research platform for cryptocurrency perpetual futures, and wrote up its results as eight working papers.

Instrumentation Rust ingestion engine subscribing to full order-book and trade streams; 236-dimensional feature vector per symbol every 100 ms across 21 categories; end-to-end feature latency below 80 ms at the 99th percentile. Write-once capture store with dual event/ingest timestamps, making lookahead a queryable property rather than a review convention.

Research layer Autonomous discovery agents under a five-gate replication protocol with Benjamini–Hochberg false-discovery-rate control and an adaptive acceptance threshold; hash-chained verdict ledger; walk-forward validation with purging and embargo. Python/Rust, roughly 4,300 automated tests, a 340-command research CLI.

Datasets held Tick-level order book, trade and funding data on BTC, ETH and SOL at 100 ms cadence; position-level liquidation data from a venue publishing per-position liquidation prices.

Working papers

All eight authored solely. [TODO: Add SSRN abstract IDs after P2; mark any arXiv posting and any venue a paper is under review at — “under review at X” is worth more than the other seven lines.]

1. *Predictive Structure of the Perpetual-Futures Limit Order Book: Universality, Orthogonality, and the Execution Boundary*. Two empirically orthogonal predictive channels; the fill-conditioning decomposition bounding monetization.
2. *Autonomous Microstructure Alpha Discovery under False-Discovery Control*. The replication apparatus, and the adverse-selection barrier as its central negative result.
3. *The Process: A First-Class Analytical Unit for Orthogonal Information Extraction*. Effective degrees of freedom of a signal book; five orthogonalisation mechanisms.
4. *Event-Aligned Singular Value Decomposition for Data-Driven Pattern Kernel Discovery*. Basis estimation provably blind to forward returns.
5. *Exact or Nothing: Gate-Governed Measurement of the Liquidation Map*. Measurement instrument; a negative result on sampling bias in liquidation data.
6. *From Liquidation Geometry to Price Movement*. Spatial density field and a closed-form cascade condition.
7. *Perception Pressure and Resonance*. Signed, decomposable media-narrative measures.
8. *From Narrative to Alpha*. Effective-breadth argument for an exogenous signal.

Technical

<i>Languages</i>	Rust, Python, C/C++ (embedded), SQL. [TODO: trim or extend to what is true]
<i>Methods</i>	Multiple-testing control (BH-FDR), walk-forward validation with purging and embargo, information-theoretic feature selection (KSG mutual information, conditional MI), random-matrix denoising, Hawkes-type intensity models, gradient boosting.
<i>Systems</i>	Real-time streaming, columnar storage (Parquet/DuckDB), containerised deployment, Prometheus/Grafana observability. [TODO: prune to what you would defend in an interview]

Languages

[TODO: e.g. Turkish (native), English (fluent), German (level) — German matters for ETH/UZH administration though not for research]

References

Available on request. [TODO: See referees.md — do not send this CV until at least two are confirmed.]